

Fall 1986

Problem 1 (Fa86). The Arzelà–Ascoli Theorem asserts that the sequence $\{f_n\}$ of continuous real valued functions on a metric space Ω is *precompact*, i.e., has a uniformly convergent subsequence, if

1. Ω is compact,
2. $\sup \|f_n\| < \infty$, where $\|f_n\| = \sup \{|f_n(x)| \mid x \in \Omega\}$,
3. the sequence is equicontinuous.

Give three examples of sequences which are not precompact, such that any of two the properties hold and the third fails. Take Ω to be a subset of the real line. Sketch the graph of a typical member of the sequence in each case.

Problem 2 (Fa86, Fa23). Let the points a , b , and c lie on the unit circle of the complex plane and satisfy $a + b + c = 0$. Prove that a , b , and c form the vertices of an equilateral triangle.

Problem 3 (Fa86). Evaluate $\iint_{\mathcal{R}} (x^3 - 3xy^2) \, dx \, dy$ where

$$\mathcal{R} = \{(x, y) \in \mathbb{R}^2 \mid (x+1)^2 + y^2 \leq 9, \quad (x-1)^2 + y^2 \geq 1\}$$

Problem 4 (Fa86, Fa13, Fa19, Fa23). Show that the polynomial $p(z) = z^5 - 6z + 3$ has five distinct complex zeros, of which exactly three (and not five) are real.

Problem 5 (Fa86). Let $M_{2 \times 2}$ denote the vector space of complex 2×2 matrices. Let

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

and let the linear transformation $T : M_{2 \times 2} \rightarrow M_{2 \times 2}$ be defined by $T(X) = XA - AX$. Find the Jordan Canonical Form for T .

Problem 6 (Fa86). Prove the following or find a counterexample: If p and q are continuous real valued functions on $\mathbb{R}$ such that $|q(x)| \leq |p(x)|$ for all x , and if every solution f of the differential equation

$$f' + qf = 0$$

satisfies $\lim_{x \rightarrow +\infty} f(x) = 0$, then every solution f of the differential equation

$$f' + pf = 0$$

satisfies $\lim_{x \rightarrow +\infty} f(x) = 0$.

Problem 7 (Fa86). Let $\mathbb{F}$ be a field containing $\mathbb{Q}$ such that $[\mathbb{F} : \mathbb{Q}] = 2$. Prove that there exists a unique integer m such that m has no multiple prime factors and $\mathbb{F}$ is isomorphic to $\mathbb{Q}(\sqrt{m})$.

Problem 8 (Fa86). Let f be a continuous real valued function on $[0, 1]$ such that, for each $x_0 \in [0, 1)$,

$$\limsup_{x \rightarrow x_0^+} \frac{f(x) - f(x_0)}{x - x_0} \geq 0$$

Prove that f is nondecreasing.

Problem 9 (Fa86). Evaluate $\int_0^\infty \frac{\log x}{(x^2 + 1)(x^2 + 4)} dx$.

Problem 10 (Fa86). For f a real valued function on the real line, define the function Δf by $\Delta f(x) = f(x + 1) - f(x)$. For $n \geq 2$, define $\Delta^n f$ recursively by $\Delta^n f = \Delta(\Delta^{n-1} f)$. Prove that $\Delta^n f = 0$ if and only if f has the form $f(x) = a_0(x) + a_1(x)x + \cdots + a_{n-1}(x)x^{n-1}$ where $a_0, a_1, \dots, a_{n-1}$ are periodic functions of period 1.

Problem 11 (Sp78, Fa82, Fa86). Let M be a matrix with entries in a field $\mathbb{F}$. The *row rank* of M over $\mathbb{F}$ is the maximal number of rows which are linearly independent (as vectors) over $\mathbb{F}$. The *column rank* is similarly defined using columns instead of rows.

1. Prove row rank = column rank.
2. Find a maximal linearly independent set of columns of

$$\begin{pmatrix} 1 & 0 & 3 & -2 \\ 2 & 1 & 2 & 0 \\ 0 & 1 & -4 & 4 \\ 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \end{pmatrix}$$

taking $\mathbb{F} = \mathbb{R}$.

3. If $\mathbb{F}$ is a subfield of $\mathbb{K}$, and M has entries in $\mathbb{F}$, how is the row rank of M over $\mathbb{F}$ related to the row rank of M over $\mathbb{K}$?

Problem 12 (Fa86). Let $\{U_1, U_2, \dots\}$ be a cover of $\mathbb{R}^n$ by open sets. Prove that there is a cover $\{V_1, V_2, \dots\}$ such that

1. $V_j \subset U_j$ for each j
2. each compact subset of $\mathbb{R}^n$ is disjoint from all but finitely many of the V_j .

Problem 13 (Fa86, Fa24). Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by:

$$f(x, y) = \begin{cases} x^{4/3} \sin(y/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Determine all points at which f is differentiable.

Problem 14 (Fa86). Let a and b be real numbers. Prove that there are two orthogonal unit vectors u and v in $\mathbb{R}^3$ such that $u = (u_1, u_2, a)$ and $v = (v_1, v_2, b)$ if and only if $a^2 + b^2 \leq 1$.

Problem 15 (Fa83, Fa86). Prove that if p is a prime number, then the polynomial

$$f(x) = x^{p-1} + x^{p-2} + \dots + 1$$

is irreducible in $\mathbb{Q}[x]$.

Problem 16 (Fa86). Discuss the solvability of the differential equation

$$(e^x \sin y)(y')^3 + (e^x \cos y)y' + e^y \tan x = 0$$

with the initial condition $y(0) = 0$. Does a solution exist in some interval about 0? If so, is it unique?

Problem 17 (Fa86). Let G be a subgroup of S_5 . Prove that if G contains a 5-cycle and a 2-cycle, then $G = S_5$.

Problem 18 (Fa86). Evaluate $\frac{1}{2\pi i} \int_{|z|=1} \frac{z^{11}}{12z^{12} - 4z^9 + 2z^6 - 4z^3 + 1} dz$ where the direction of integration is counterclockwise.

Problem 19 (Fa86). Prove that if six people are riding together in an Evans Hall elevator, there is either a three-person subset of mutual friends (each knows the other two) or a three-person subset of mutual strangers (each knows neither of the other two).

Problem 20 (Fa86). Let f be a real valued continuous function on $[0, \infty)$ such that

$$\lim_{x \rightarrow \infty} \left(f(x) + \int_0^x f(t) \, dt \right)$$

exists. Prove that

$$\lim_{x \rightarrow \infty} f(x) = 0$$